Writer Monad

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Produce a value and an accumulated log.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Lab notebook beside the experiment.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.writer;

import com.buildwithwaqar.patterns.functional.monoid.Monoid;

import java.util.function.Function;

public record Writer<W, A>(W log, A value) {
    public <B> Writer<W, B> map(Function<? super A, ? extends B> f) {
        return new Writer<>(log, f.apply(value));
    }

    public <B> Writer<W, B> flatMap(Monoid<W> monoid, Function<? super A, Writer<W, B>> f) {
        var next = f.apply(value);
        return new Writer<>(monoid.combine(log, next.log), next.value);
    }

    public static <W, A> Writer<W, A> pure(Monoid<W> monoid, A a) {
        return new Writer<>(monoid.empty(), a);
    }

    public static <W> Writer<W, Void> tell(W w) { return new Writer<>(w, null); }
}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/writer/
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick Writer Monad? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Produce a value and an accumulated log.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Lab notebook beside the experiment. [0:40] Intent — State the design force in one sentence: Produce ...